Section A

Answer **all** questions in this section in the spaces provided.

1 Compound **X** $\xrightarrow{\text{heat}}$ Compound **Y** $\xrightarrow{\text{heat}}$ Compound **Z**

	Compound **X**	Compound **Y**	Compound **Z**
% by mass	oxygen 13.38% metal **M** 86.62%	oxygen 9.33% metal **M** 90.67%	oxygen 7.17% metal **M** 92.83%

Compound **X** is a dark brown compound which consists of oxygen and metal **M** only. Successive heating of compound **X** transformed it into two other compounds.

(a) Calculate the mass of oxygen that is combined with 1.00 g of metal **M** in each of these three compounds.

In compound **X**, 1.00 g metal **M** combines with ………………… g of oxygen.

In compound **Y**, 1.00 g metal **M** combines with ………………… g of oxygen.

In compound **Z**, 1.00 g metal **M** combines with ………………… g of oxygen.

[2]

(b) The empirical formula of compound **X** is **M**O_2.
By considering the ratios of the amount of oxygen present in the three compounds, Determine the empirical formulae for compound **Y** and compound **Z.**

Compound **Y** ....

Compound **Z** ....

[3]

(c) Identify metal **M** by calculating its molar mass.

Metal **M** is .... with a molar mass of .... g mol^{-1}.

[2]

[Total: 7]

2 This question is about halogens and its compounds.

(a) World War I is sometimes known as 'The Chemists' War'. Knowledge of Chemistry was applied towards developing high explosives and new methods such as the large scale use of poison gas.

The first successful use of chlorine as a poison gas was at Ypres, Belgium, on 22 April 1915. 170 tonnes (1 tonne = 1000 kg) of chlorine contained in 5730 cylinders was released forming a grey-green cloud which drifted across French troops. Chlorine can damage the eyes, nose, throat and lungs and is fatal at concentrations of 1000 ppm and above (1 ppm = 1 mg dm^{-3}). Early counter-measures to chlorine included instructing troops to cover their mouths with gauze pads soaked in sodium hydrogen carbonate solution. Eventually, more effective counter-measures to chlorine were developed and thus other poison gases were introduced.

(i) Calculate the maximum amount of chlorine gas that could be released from one of the cylinders that was used at Ypres on 22 April 1015.

(ii) Determine the concentration of chlorine gas, in mol dm^{-3}, in 1000 ppm of the gas.

(iii) In an accident, the chlorine gas from one such cylinder was released into a factory room of volume 25.0 m^3.

Determine if the concentration of chlorine gas was fatal. Assume that the gas was released at room temperature and pressure.

[4]

(b) Strontium chloride is used to prepare strontium ethanedioate, SrC_2O_4, which is widely used as a red colorant in fireworks. The equation for the reaction is as shown:

$$SrCl_2(aq) + Na_2C_2O_4(aq) \rightarrow SrC_2O_4(s) + 2NaCl(aq)$$

(i) State the type of bonding present in SrC_2O_4.

....

(ii) Deduce the shape and bond angle in the ethanedioate part of SrC_2O_4 by drawing its structural formula.

(iii) State the number of sigma and pi bonds present in the ethanedioate part of SrC_2O_4.

....

[4]

(c) **(i)** Halogens can react with aluminium to form halides such as AlF_3 and $AlBr_3$. When heated, aluminium bromide remains as a solid until 371K whereas aluminium fluoride remains as a solid until 1530K.

Account for the difference in their melting points in terms of structure and bonding.

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(ii) In the vapour phase, an equilibrium is established between aluminium bromide and its dimer as follows:

$$2AlBr_3(g) \rightleftharpoons Al_2Br_6(g)$$

With the help of a dot-and-cross diagram, explain how the dimer is formed.

[5]

[Total: 13]

3 This question is about Period 3 elements and their compounds.

(a) The ionic radii of four consecutive elements in period 3 are given below.

element	**A**	**B**	**C**	**D**
ionic radius / nm	0.065	0.050	0.041	0.212

(i) Deduce which one of the elements, **A**, **B**, **C** or **D** corresponds to magnesium. Explain your answer.

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(ii) Period 3 elements react with chlorine to form chlorides. Account for the difference in the pH values when chloride of **B** and chloride of **D** is separately dissolved in water. Write equations to support your answer.

[6]

(b) Period 3 elements also form oxides upon heating with oxygen.

(i) Write equations to illustrate the reaction between sodium oxide and water as well as phosphorous pentoxide and water. Hence, deduce the pH of the resultant solution after mixing. Support your answers with relevant equations.

(ii) Write equations to illustrate the amphoteric behaviour of aluminium oxide, using potassium hydroxide and sulfuric acid in your answer.

[4]

[Total: 10]

4 **(a)** Organic halogen compounds have a wide variety of commercial uses. For example, the polymer, polyvinyl chloride (PVC) is commonly used as insulation for electrical wiring.
The monomer of PVC, $CH_2{=}CHCl$, is industrially produced from $CH_2{=}CH_2$ via the following route:

$$CH_2{=}CH_2 \xrightarrow{\text{Step I}} CH_2ClCH_2Cl \xrightarrow{\text{Step II}} CH_2{=}CHCl$$

(i) Step **I** is carried out by reacting $CH_2{=}CH_2$ with hydrogen chloride and oxygen over a copper(II) chloride catalyst. Write a balanced equation for this reaction.

....

(ii) Suggest the reagents and conditions that can be used to carry out step **II** in the laboratory.

....
[2]

(b) **(i)** $CH_2{=}CHCl$ can be formed directly in a single step from $CH_2{=}CH_2$ in the following way:

$$CH_2{=}CH_2 \xrightarrow[\text{uv light}]{Cl_2} CH_2{=}CHCl$$

This reaction also results in the formation of a product, $C_2H_2Cl_2$, which exists as a pair of geometric isomers **R** and **S**. Give the name of this product.

....

(ii) Some physical properties of the geometric isomers, **R** and **S** are shown below:

Isomer	Melting point / °C	Boiling point / °C
R	–49.4	48.5
S	–81.5	60.2

Draw the structure of **R** and **S** in the boxes below.

R	**S**

(iii) $CH_3CH_2CH_3$ can also react with Cl_2 in the presence of uv light to form 2 monochlorinated products, $CH_2ClCH_2CH_3$ and $CH_3CHClCH_3$. It has been determined experimentally that the hydrogen atoms in alkanes are replaced by chlorine atoms at different rates as shown below:

Type of H atom	Reaction	Relative rate of replacement
Primary	$RCH_3 \rightarrow RCH_2Cl$	1
Secondary	$R_2CH_2 \rightarrow R_2CHCl$	3.5
Tertiary	$R_3CH \rightarrow R_3CCl$	5

–R represents an alkyl group.

By considering both the number and the rate of replacement of each type of hydrogen atom present in $CH_3CH_2CH_3$, predict the ratio of $CH_2ClCH_2CH_3$ and $CH_3CHClCH_3$ formed. Show your working clearly.

[4]

(c) Chlorofluorocarbons (CFCs) are another class of organic halogen compounds. They were formerly used in aerosol propellants but due to concerns about their effect on the depletion of the ozone layer, they have been replaced by volatile hydrocarbons such as $CH_3CH_2CH_3$ and $CH_3CH_2CH_2CH_3$.

Two common CFCs once used in aerosol propellants were CCl_3F and CH_3CClF_2.

(i) By comparing the strength of the relevant carbon–halogen bonds, suggest and explain which compound, CCl_3F or CH_3CClF_2, is likely to have a smaller effect on ozone depletion.

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(ii) Other than aerosol propellants, give one use of CFCs.

....

....

(iii) State one possible hazard of using hydrocarbons instead of CFCs in aerosol propellants.

....

[4]

[Total: 10]

Section B

Answer **two** questions from this section on separate answer paper.

5 Hydrogen has large scale use in the chemical industry. Although hydrogen is the most abundant element in the Universe, very small quantities of molecular hydrogen have been found to occur naturally.

(a) One of the most efficient methods of producing hydrogen involves steam reforming of natural gas, an equilibrium reaction in which methane gas is reacted with steam to produce carbon monoxide and hydrogen.

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$$

(i) Use appropriate bond energies given in the *Data Booklet* to calculate an approximate value for the steam reforming of natural gas.
[Take the bond energy for the $C{\equiv}O$ bond in CO to be 1077 kJ mol^{-1}]

(ii) Use the energy cycle and data given below to calculate another approximate value for the steam reforming of natural gas.

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$$

$$\downarrow$$

$$CO_2(g) + 3H_2O(l)$$

$\Delta H_c^{\ominus}$ of $CH_4(g)$ = –890 kJ mol^{-1}
ΔH_{vap} of $H_2O(l)$ = +40.7 kJ mol^{-1}
$\Delta H_c^{\ominus}$ of $CO(g)$ = –283 kJ mol^{-1}
$\Delta H_f^{\ominus}$ of $H_2O(l)$ = –286 kJ mol^{-1}

(iii) Suggest two reasons for the discrepancy between the values in **(i)** and **(ii)**.

[6]

(b) The water–gas shift reaction is often used together with the steam reforming of natural gas to produce more hydrogen. This involves the highly exothermic reaction of carbon monoxide with steam to produce carbon dioxide and hydrogen.

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

The equilibrium constant, K_c is found to be 0.64 at 1000 K.

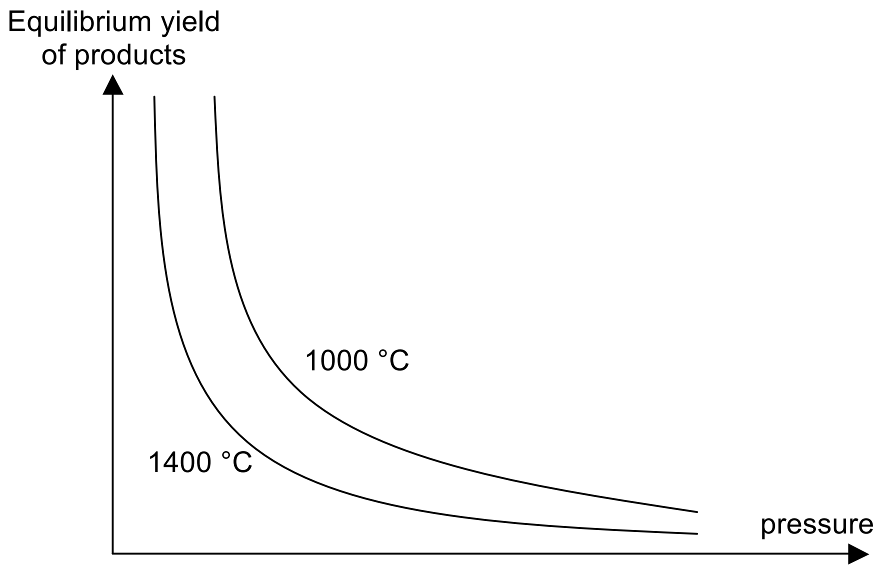
(i) Write an expression for K_c of this reaction.

(ii) A mixture containing 0.80 mol of CO, 0.80 mol of H_2O, 0.40 mol of CO_2 and 0.40 mol of H_2 was placed in a 2 dm^3 flask and allowed to come to equilibrium at 1000 K.

Calculate the amount, in mol, of each substance present in the equilibrium mixture at 1000 K.

[5]

(c) The graph below shows the variation in equilibrium yield of the products with temperature and pressure for the water–gas shift reaction.



(i) State *Le Chatelier's Principle*.

(ii) Hence comment on the accuracy of the graph with respect to both temperature and pressure.

[5]

(d) In the presence of a catalyst mixture made up of alumina, copper and zinc oxide, carbon monoxide and hydrogen reacts to form compound **D** instead.

Compound **D** is commonly obtained from biomass and consist only of the elements C, H and O. When 0.55 g of **D** was completely burnt in air, the heat produced raised the temperature of 100 g of water by 29.4 °C. The standard enthalpy change of combustion of **D** is found to be –715 kJ mol^{-1}.

(i) Calculate the amount of heat released in the experiment.
[Assume the heat capacity of all solutions to be 4.18 J K^{-1} cm^{-3}]

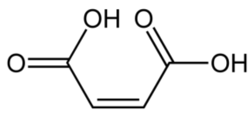
(ii) Hence deduce the molecular formula of compound **D**.

[4]

[Total:20]

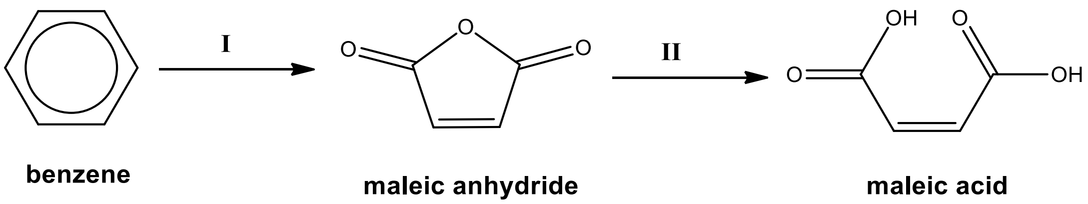
6 “The bubble has burst for several tea businesses here” quoted from The Straits Times on 28 May 2013.

The Agri-Food and Veterinary Authority of Singapore (AVA) recalled 11 types of tapioca starch balls - also known as "pearls" - made in Taiwan that are used in popular 'bubble teas', after they were found to contain maleic acid. Consuming high levels of maleic acid can cause kidney damage.



Maleic acid

(a) Maleic acid is commonly produced via the following 2-step synthetic route industrially.



(i) State the type of reactions for **I** and **II**.

(ii) Write a balanced equation for step **II**.

(iii) Suggest how you can distinguish the following pairs of compounds:
I – Benzene and maleic anhydride
II – Maleic anhydride and maleic acid

In each case, state the reagents and conditions as well as observations for each compound.

[7]

(b) Draw the structure of the major product formed when maleic acid reacts with

(i) cold concentrated sulfuric acid, followed by hydrolysis

(ii) hydrogen gas with palladium on carbon

(iii) thionyl chloride

(iv) methanol with concentrated sulfuric acid

(v) cold alkaline potassium manganate(VII)

[6]

(c) **(i)** State the type of isomerism displayed by maleic acid and draw its isomer, fumaric acid.

(ii) Based on the structures of maleic acid and fumaric acid, predict which acid has a lower melting point.

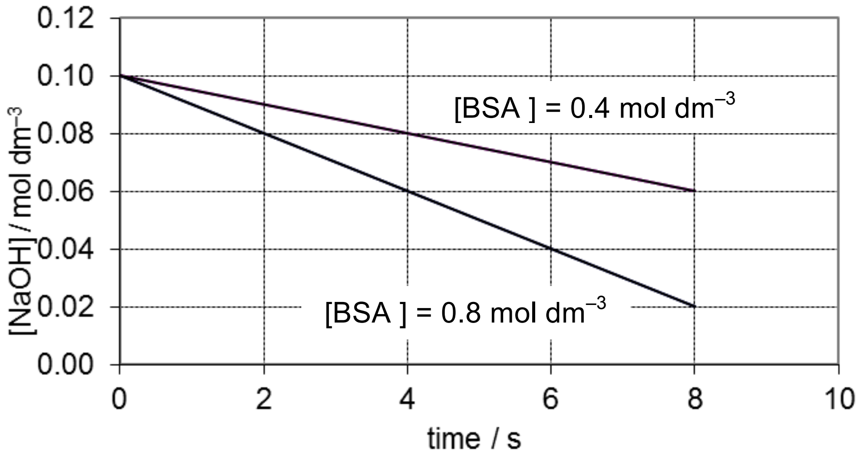
(iii) A student added some water to a mixture of maleic acid and fumaric acid and performed a simple filtration. He concluded that the residue obtained must be fumaric acid.

Do you agree with the student’s statement? Justify your answer.

[7]

[Total: 20]

7 (a) Benzenesulfonic acid (BSA) reacts with aqueous NaOH to produce phenol. The graph below shows the reaction between aqueous sodium hydroxide and solutions of benzenesulfonic acid of different concentration at temperature T ^{o}C.

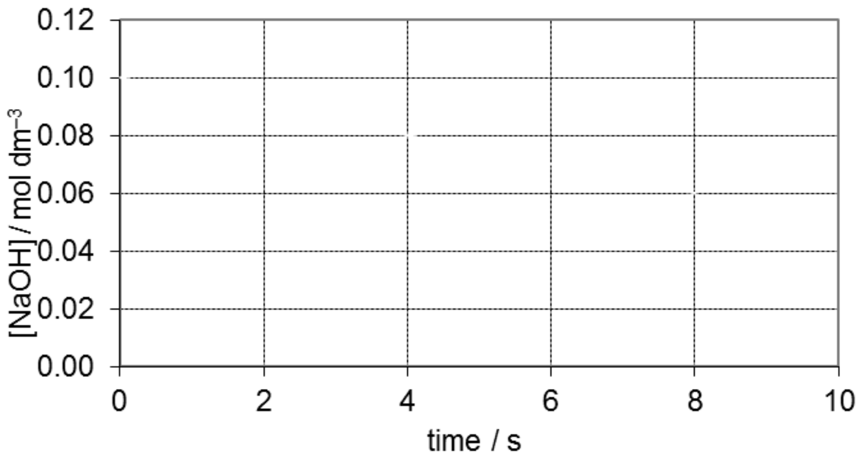


(i) With reference to the graph above, deduce the order of reaction with respect to sodium hydroxide and benzenesulfonic acid and hence write down the rate equation.

(ii) Suggest a method to monitor the concentration of NaOH in the above kinetic study.

(iii) Determine the time taken for the concentration of benzenesulfonic acid to reduce from 0.4 mol dm^{-3} to half of its original concentration.

(iv) On the grid drawn, sketch and label a graph for the reaction of 0.10 mol dm^{-3} of aqueous sodium hydroxide with 0.4 mol dm^{-3} of BSA at (T−10) ^{o}C.



(v) Account for the shape of the graph in part **(iv)**.

[9]

(b) Hypochlorous acid, HC*l*O, is a *weak acid* that is formed when chlorine reacts with water.

(i) With the aid of an equation, explain what is meant by weak acid.

(ii) Write an expression for the acid dissociation constant, K_a, for HC*l*O.

(iii) The following equilibrium concentrations were determined for a solution of HC*l*O, calculate the value of K_a of HC*l*O, stating the units.

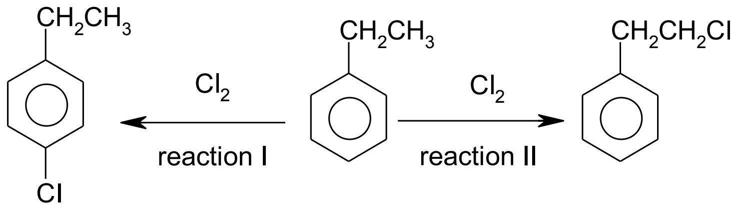
Species	Concentration / mol dm^{-3}
HC*l*O(aq)	0.0200
H^+(aq)	2.50 x 10^{-5}
C*l*O^-(aq)	2.50 x 10^{-5}

[3]

(c) A solution containing hypochlorous acid, HC*l*O, and sodium chlorate(I), NaC*l*O, can act as a buffer solution. Explain, with the help of equations, how this solution can regulate pH when relatively small amount of acid or base is added to the solution.

[3]

(d) Ethylbenzene can react with chlorine in two ways, depending on the conditions of the reaction.



A **B**

(i) State the condition needed for reaction **I**.

(ii) Describe a chemical test to distinguish between **A** and **B**. State the reagents, conditions, and the observations for each compound. Write balanced equations for any reactions that occur.

[5]

[Total: 20]